MULTIMEDIA UNIVERSITY OF KENYA

Bridging the Informal Labour Market

A Web Platform for Blue-Collar Service Connectivity in Kenya

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Background of study

A Vital, Yet Unconnected Workforce

According to the Kenya National Bureau of Statistics (KNBS, 2023), the informal sector drives over **80 percent** of total employment in Kenya.

Millions of skilled tradespeople operate entirely within this segment without structural visibility or verified digital channels.

This massive sector remains underserved by digital transformation, relying heavily on traditional word-of-mouth referrals that are geographically constrained.

Problem Statement

The Worker Gaps

Reliance on word-of-mouth referrals leads to **persistent underemployment**, severe income instability, and prolonged periods without reliable service requests.

Highly talented independent artisans remain digitally invisible to broader, paying local markets outside their immediate geographic circles.

The Client Gaps

Clients face a persistent **trust deficit** with no standard system to verify technical qualifications, historic pricing, or reliability.

Hiring remains an uncertain process, exposing households and businesses to quality issues, project delays, and financial overcharges.

Aim of the Study

🎯 Core Mission

The primary aim is to **design, develop, and evaluate** an intuitive mobile-friendly platform connecting independent Kenyan blue-collar workers with active local clients.

🔄 Structural Shift

By replacing insecure, informal hiring loops with verified digital tools, the platform modernizes access, ensures work consistency, and builds high-trust local micro-economies.

Research Objectives

1. Analyze Gaps

Identify the core visibility, workflow and trust bottlenecks limiting local artisan hiring and service discovery.

2. Build System

Design a simplified mobile platform workflow enabling clean profiles, reliable listings, and visual work portfolios.

3. Validate Trust

Develop trusted verification pipelines, real-time booking coordinates, and moderated rating matrices.

Significance of the Study

- **Enhanced Employment Access:** Equips underemployed artisans with digital visibility to secure predictable income and expand their trade presence.
- **Trust-Centric Marketplace:** Minimizes safety and verification concerns for households through transparent portfolios and performance ratings.
- **Sustainable Economic Growth:** Champions formalization of informal markets, building micro-entrepreneurship loops in Kenya's primary labor engine.

Scope and Assumptions

System Boundaries

Focused on individual trades (plumbers, electricians, mechanics, cleaners) and household clients. Features include authentication, portfolio uploads, direct pricing negotiations, and verified ratings.

Key Project Assumptions

Assumes that independent artisans have access to basic internet-enabled smartphones, possess minimal mobile app navigation literacy, and provide authentic registration credentials.

Limitations of the Study

- **Diverse Digital Literacy:** Some target skilled workers may initially struggle with profile configuration, portfolio setup, or active job request processing.
- **Intermittent Local Connectivity:** Weak mobile networks in peri-urban areas can result in delayed chat messaging, notifications, or booking updates.
- **Verification and Identity Gaps:** Verifying technical capabilities and past offline claims without centralized official professional guilds.

Related Systems

Jobzy Kenya

Strengths: Targets gig matching across informal roles (cleaning, tutoring) and attempts to offer parallel financial savings modules.

Weaknesses: Highly concentrated in core urban zones with complex app navigation and high-bandwidth requirements.

Jiji Kenya (Services)

Strengths: Popular classified system providing wide advertising reach and simple photo attachment listings.

Weaknesses: Non-curated approach lacks quality audits, standardized job tracking, reviews, or secure negotiation logs.

Selected Methodology: Agile

Iterative Design Cycles

An incremental, sprint-driven approach was prioritized over Waterfall to allow ongoing feedback loops with real-world workers and supervisors.

Adaptive Sprints: Enables rapid MVP rollouts, continuous usability evaluation, and flexible requirements modeling as platform features are validated in the field.

Data Collection Methods

1. Questionnaires

Digital surveys via Google Forms focused on capture rates, pricing challenges, and mobile app usage metrics.

2. Interviews

Semi-structured discussions outlining specific trust barriers, rating confidence, and platform design expectations.

3. Field Observations

On-site tracking of real-world interactions, booking flows, and payment patterns within dense urban spaces.

Project Budget

Resource Category	Resource Description & Justification	Estimated Cost (KES)
Hardware Requirements	Workstation laptop and diverse mobile test units	80,000 KES
Software & Cloud tools	UX wireframing, PostgreSQL DB licenses, cloud hosting	25,000 KES
Network Connectivity	Mobile data and high-speed broadband lines	8,000 KES
Participant Incentives	Token compensations for field user testing groups	8,000 KES
Project Documentation	Thesis printing, technical literature, transport	5,000 KES

Functional Requirements

- **Identity Administration:** Separate workflows for secure customer and worker authentication utilizing modern JWT-based logins.
- **Dynamic Profile Building:** Artisan hubs capturing validated skill categories, localized availability, and offline porting tools.
- **Discovery & Location Match:** Advanced search models matching trade needs (e.g., plumbing) with closest available technicians.
- **Transparent Project Cycles:** Direct chat negotiations, progress updates, and ratings logged strictly post-completion.

Non-Functional Requirements

1. Accessibility

Extremely simplified, high-contrast, visual-centric UI designed for users of varying levels of digital literacy.

2. Performance

Low-bandwidth optimization and image caching to load matching results within 2-3 seconds on slow mobile networks.

3. Security

Robust password hashing, encrypted transit channels, and strict PostgreSQL transaction logs.

Image Sources

https://media.istockphoto.com/id/1418155799/photo/male-plumber-engineer-working-at-sewer-pipes-area-at-construction-site-african-american-male.jpg?s=612x612&w=0&k=20&c=xfGq5YtQOsH0StHUehCBQcU9IY2PDwqCHWtcbs7oIDA=

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